

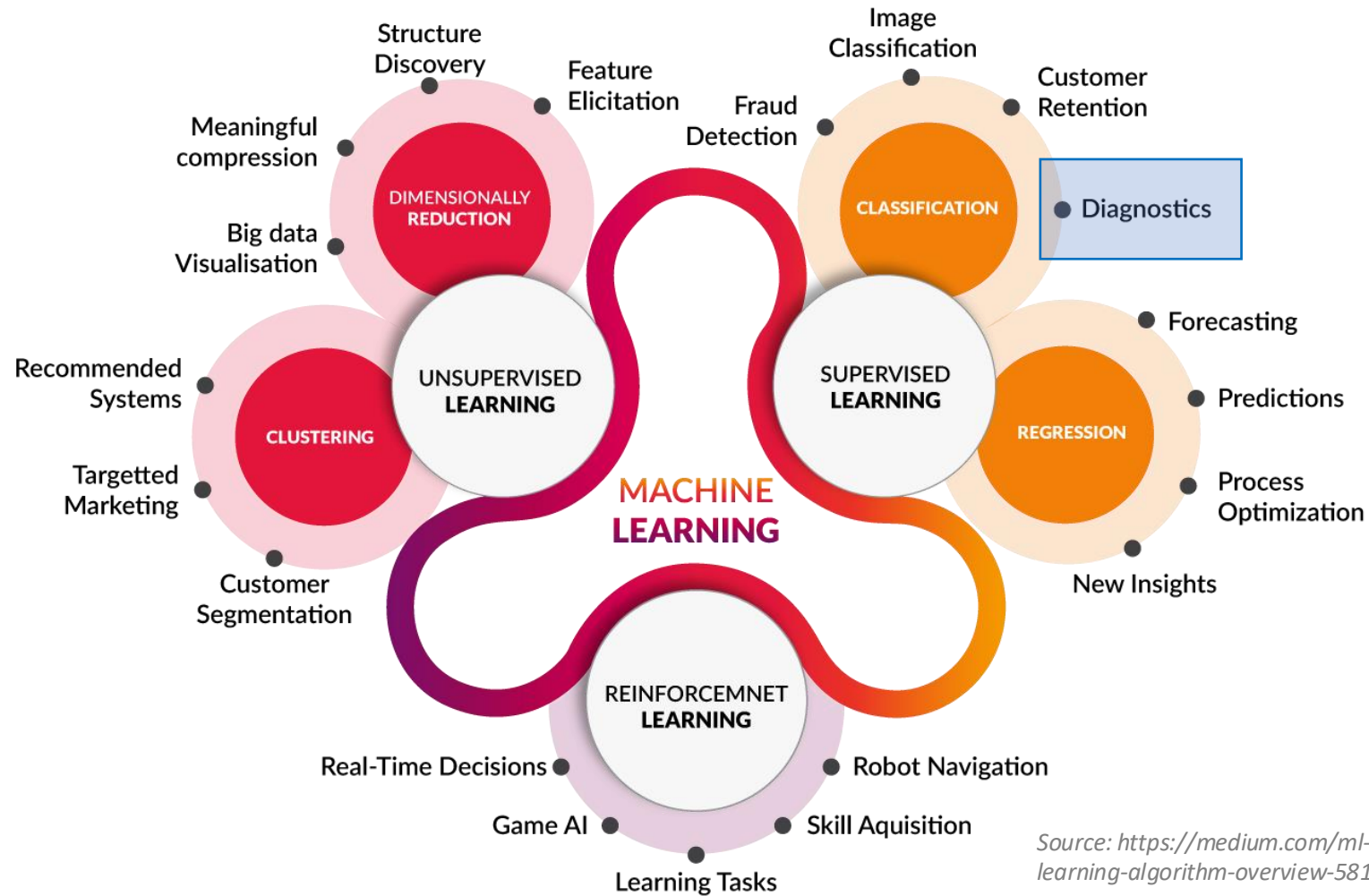
# ME597: Machine Learning (ML) Tutorial 5

Week 4

School of Mechanical Engineering  
Purdue University

Dr. Martin Jun

# Machine Learning Applications



Source: <https://medium.com/ml-research-lab/machine-learning-algorithm-overview-5816a2e6303>

# Machine Learning Algorithms

## Machine Learning

### Supervised Learning

- Regression (ML 5)
- **ANN (Artificial Neural Network)** → **Deep Learning**
- ARIMA, Exponential Smoothing  
→ Mainly for prediction (also classification)
- KNN (K-Nearest Neighbor)
- SVM (Support Vector Machine)
- Decision Tree
- Random Forest  
→ Mainly for classification (also prediction)

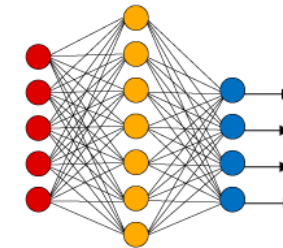
### Unsupervised Learning

- K-means Clustering
- Density-based Clustering
- Fuzzy Clustering  
→ Mainly for clustering (classification without labels)
- Association Rule Analysis
- Network Analysis  
→ Mainly for data structure/pattern recognition
- PCA(Principal Component Analysis)
- t-SNE(Stochastic Neighbor Embedding) ] (ML 4)  
→ Mainly for dimensionality reduction (visualization)

### Deep Learning

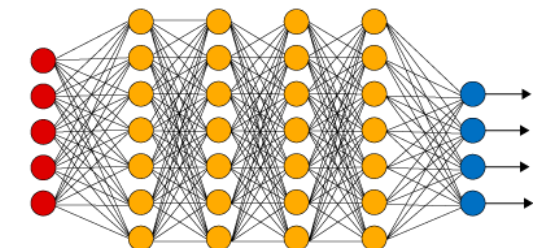
- **CNN (Convolutional Neural Network)** (ML 6)
- **RNN (Recurrent Neural Network)**
- **AE (Autoencoder)**
- GAN (Generative Adversarial Network)
- Transformer
- ...
- Generative Pre-trained Transformer (GPT)

Simple Neural Network



● Input Layer

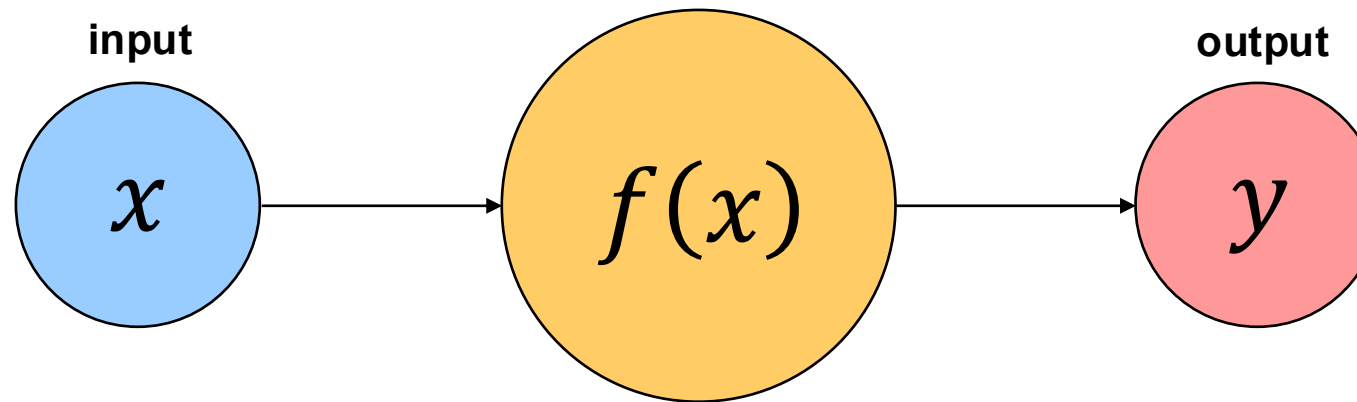
Deep Learning Neural Network



● Hidden Layer

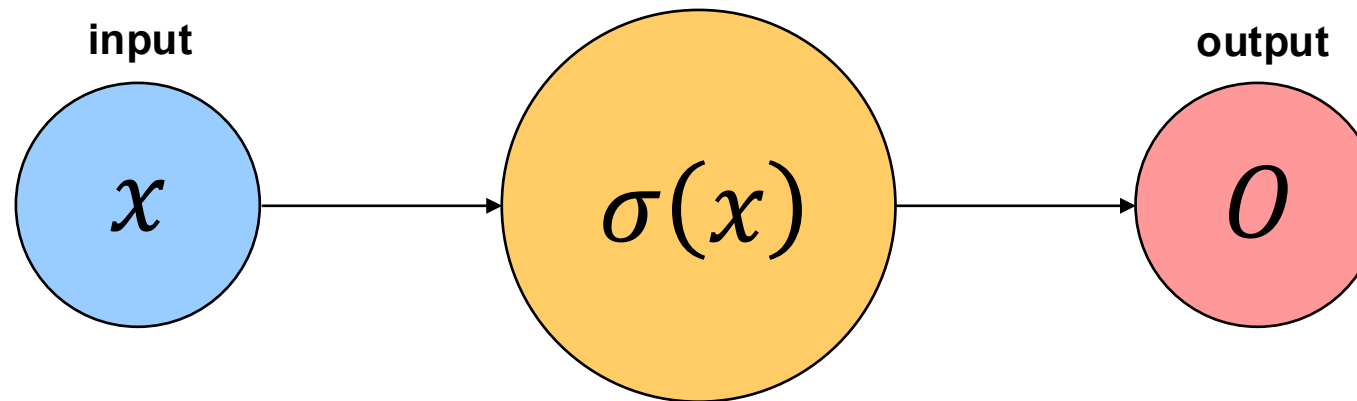
● Output Layer

# Artificial Neural Network (ANN)



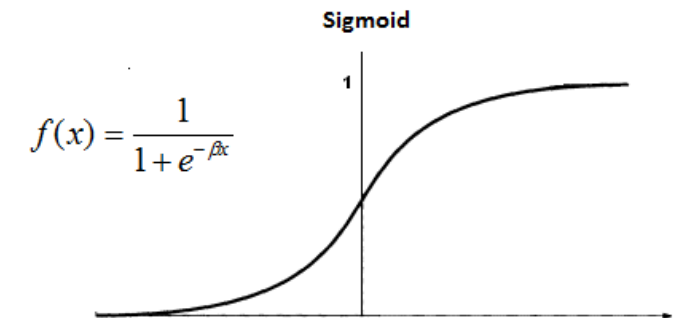
$$y = f(x) = 2x$$

# Artificial Neural Network (ANN)

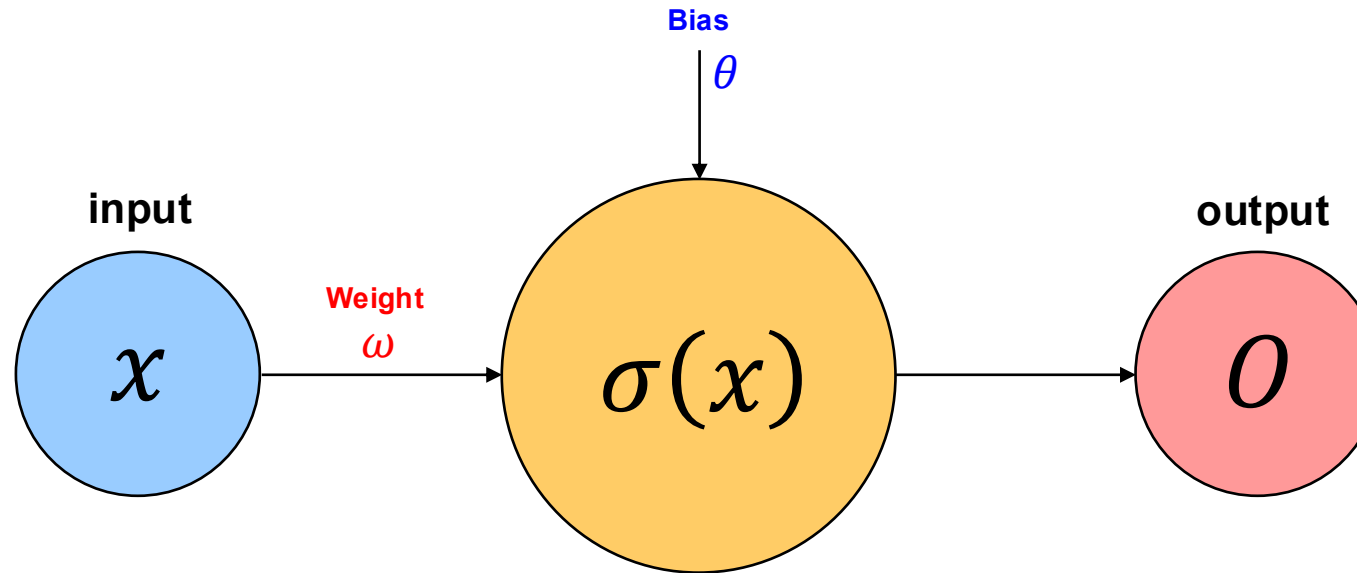


$$O = \sigma(x) = \frac{1}{1 + e^{-x}}$$

Sigmoid function

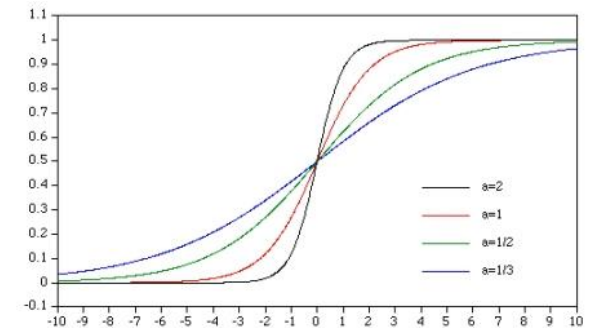


# Artificial Neural Network (ANN)

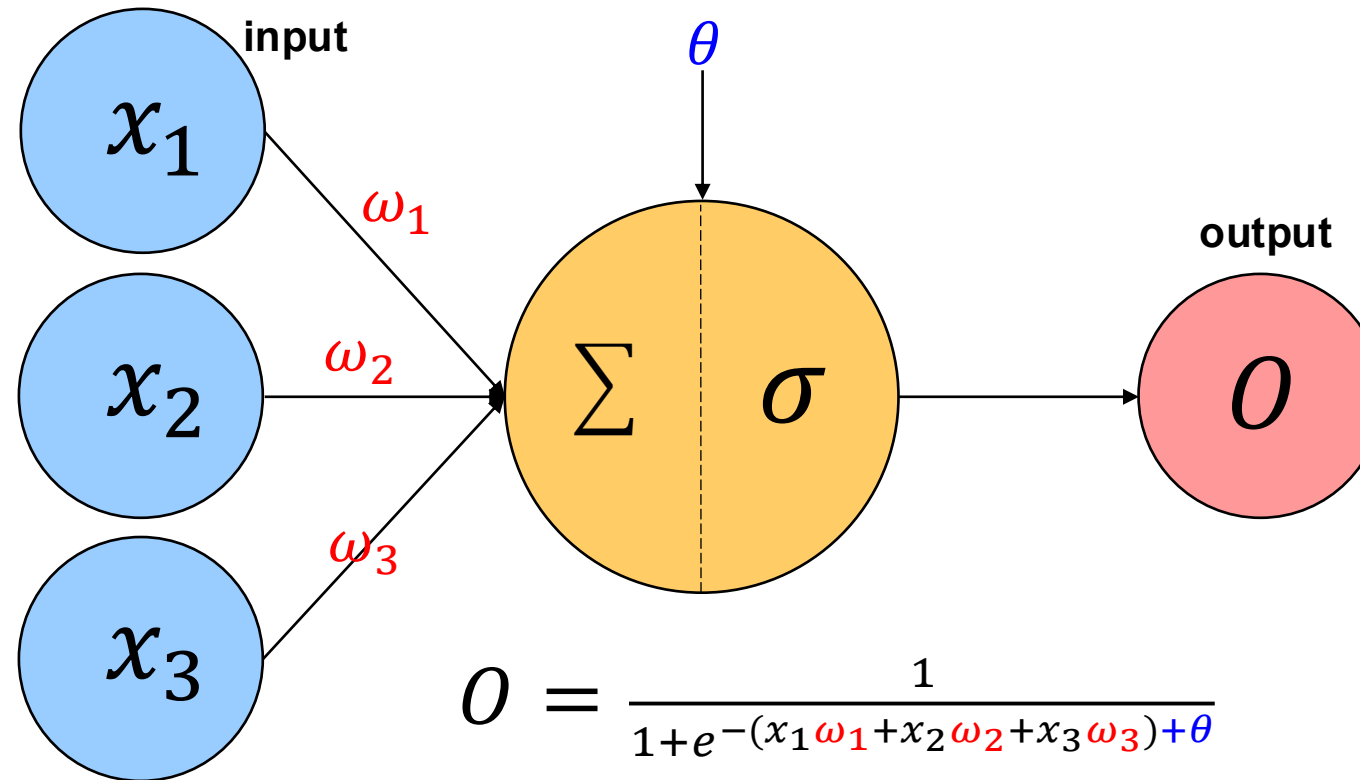


$$O = \sigma(x) = \frac{1}{1 + e^{-x\omega + \theta}}$$

Change according to  $\omega$

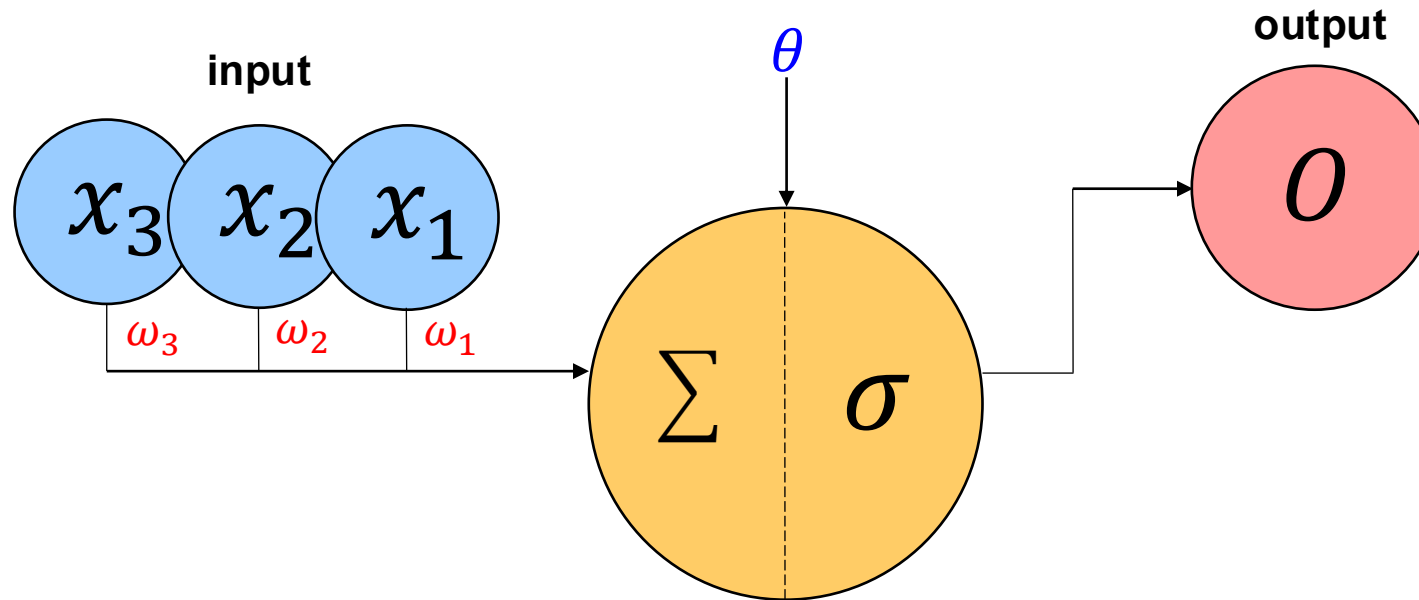


# Artificial Neural Network (ANN)



**Perceptron**

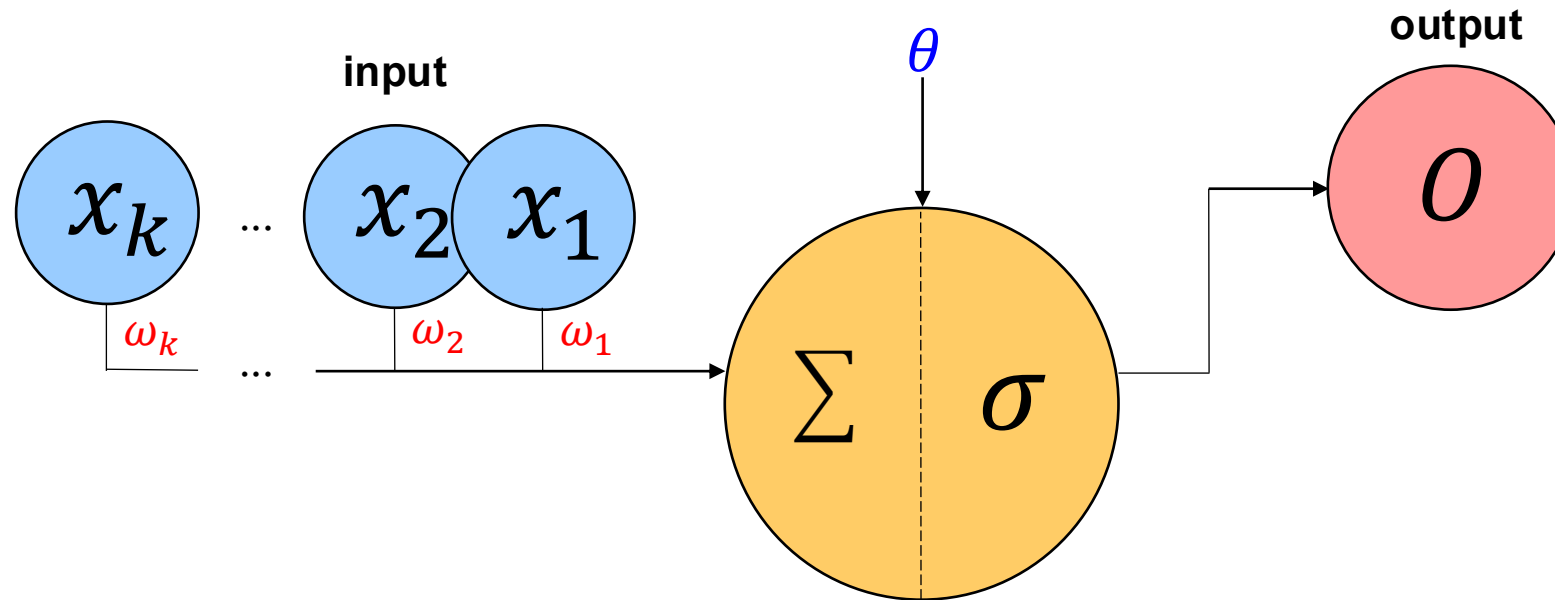
# Artificial Neural Network (ANN)



$$0 = \frac{1}{1 + e^{-(x_1 \omega_1 + x_2 \omega_2 + x_3 \omega_3) + \theta}}$$

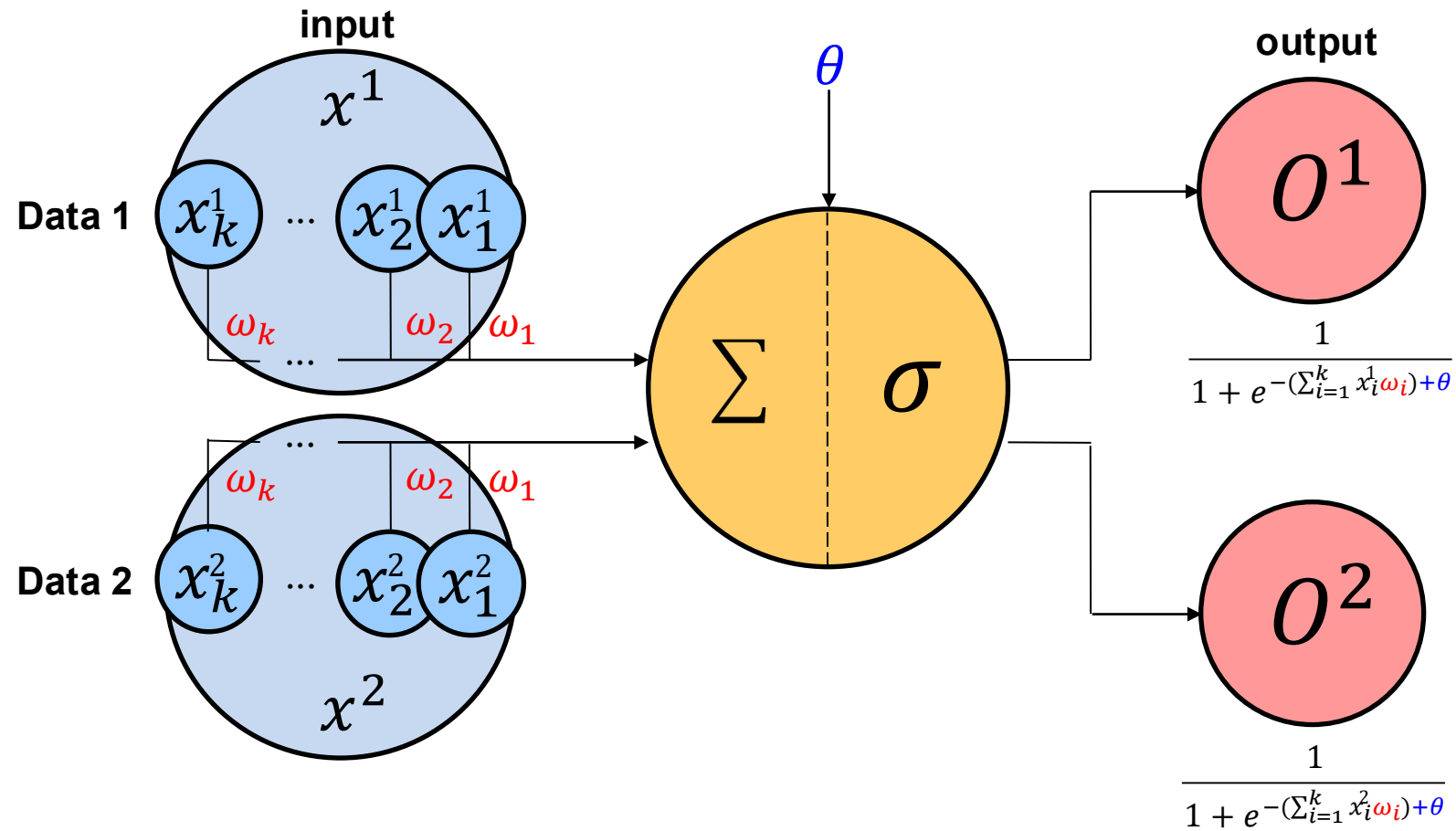


# Artificial Neural Network (ANN)

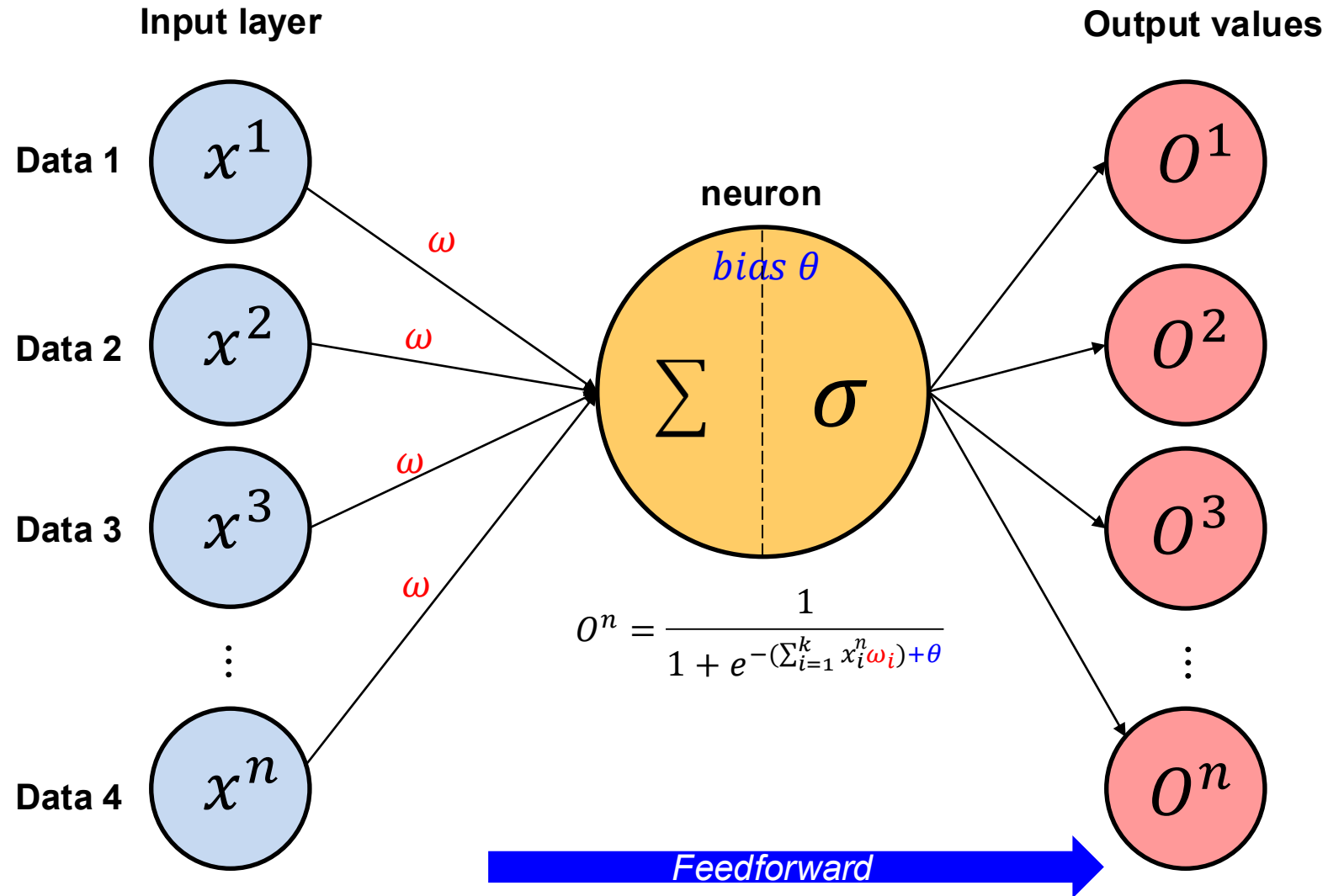


$$O = \frac{1}{1 + e^{-(\sum_{i=1}^k x_i \omega_i) + \theta}}$$

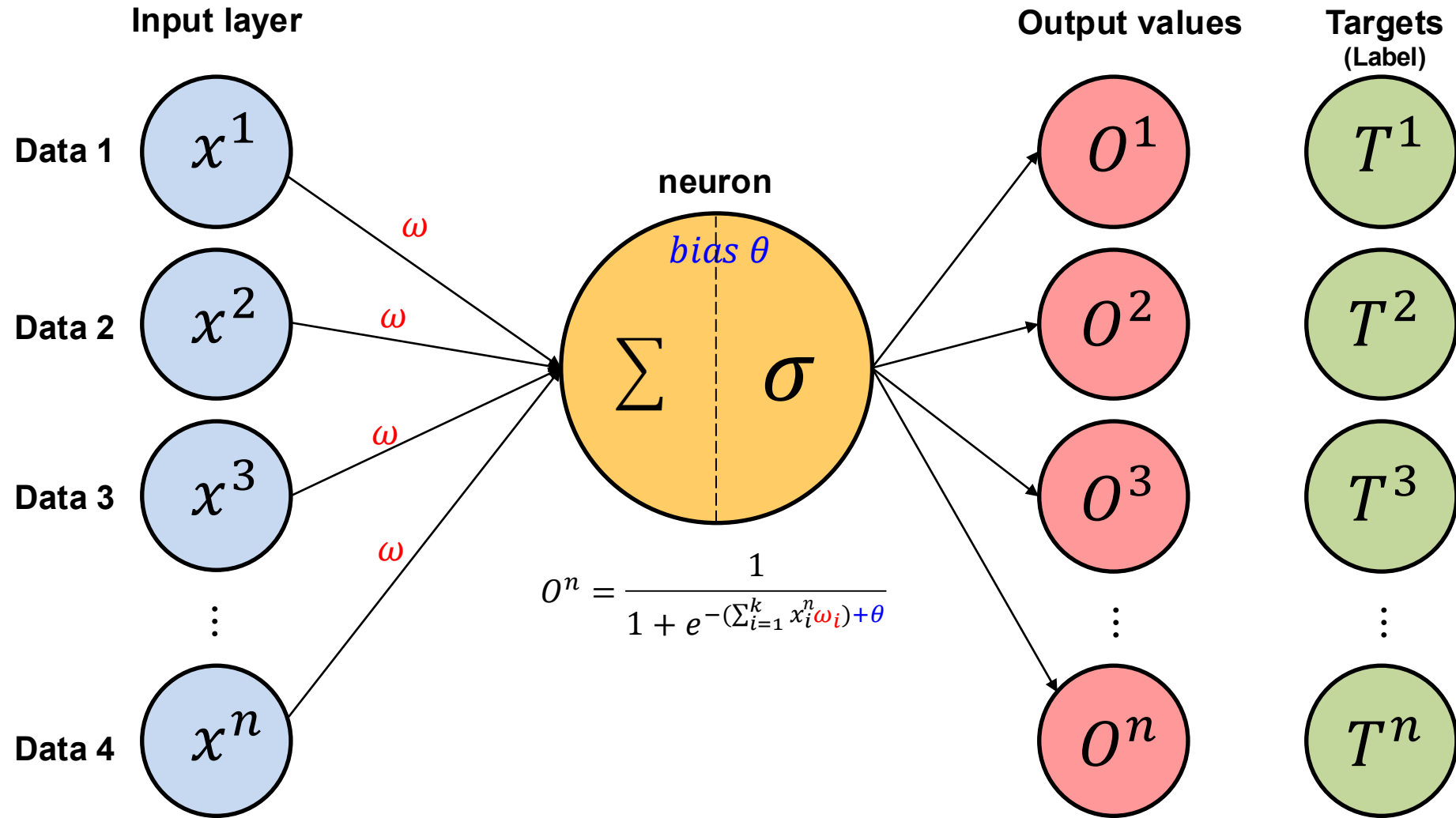
# Artificial Neural Network (ANN)



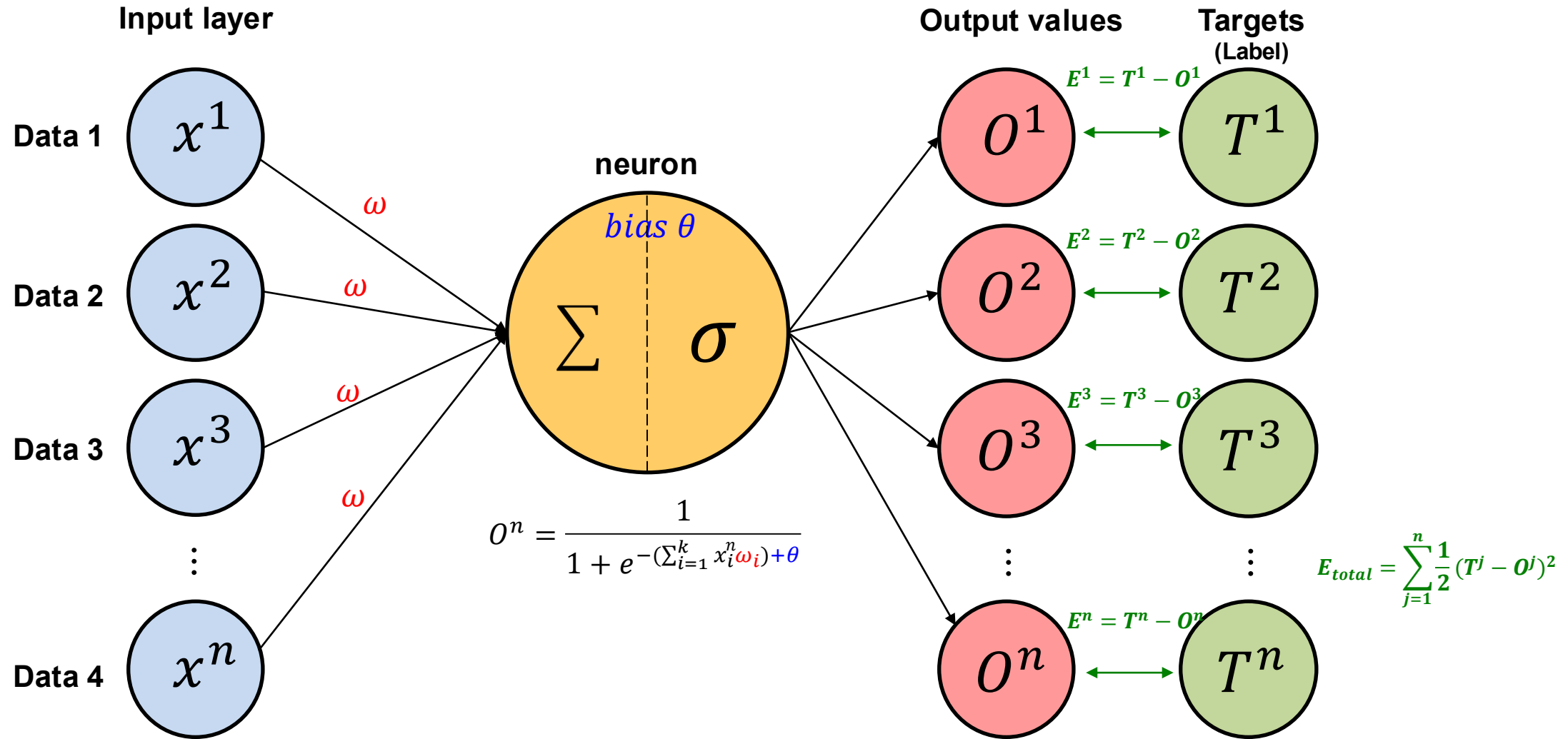
# Artificial Neural Network (ANN)



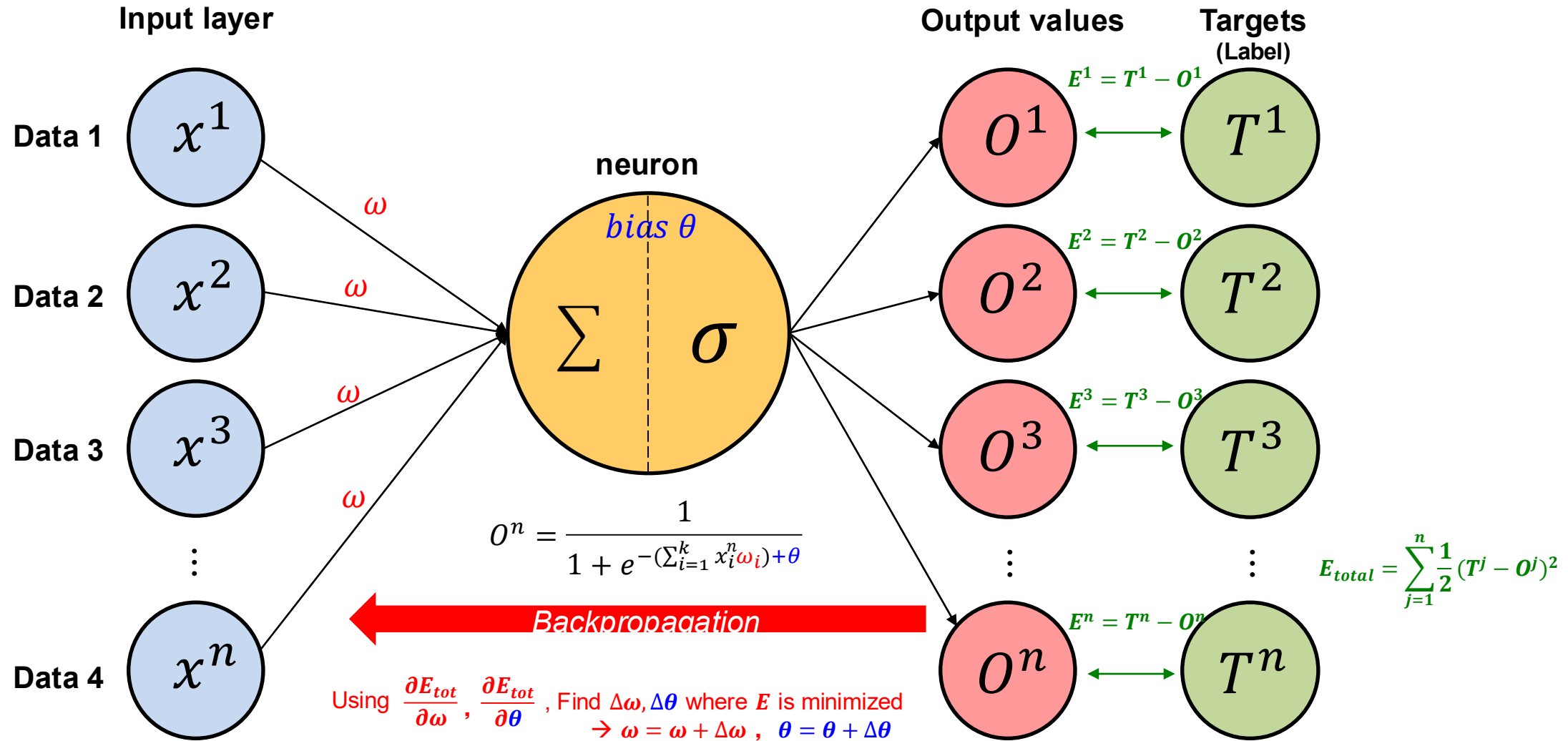
# Artificial Neural Network (ANN)



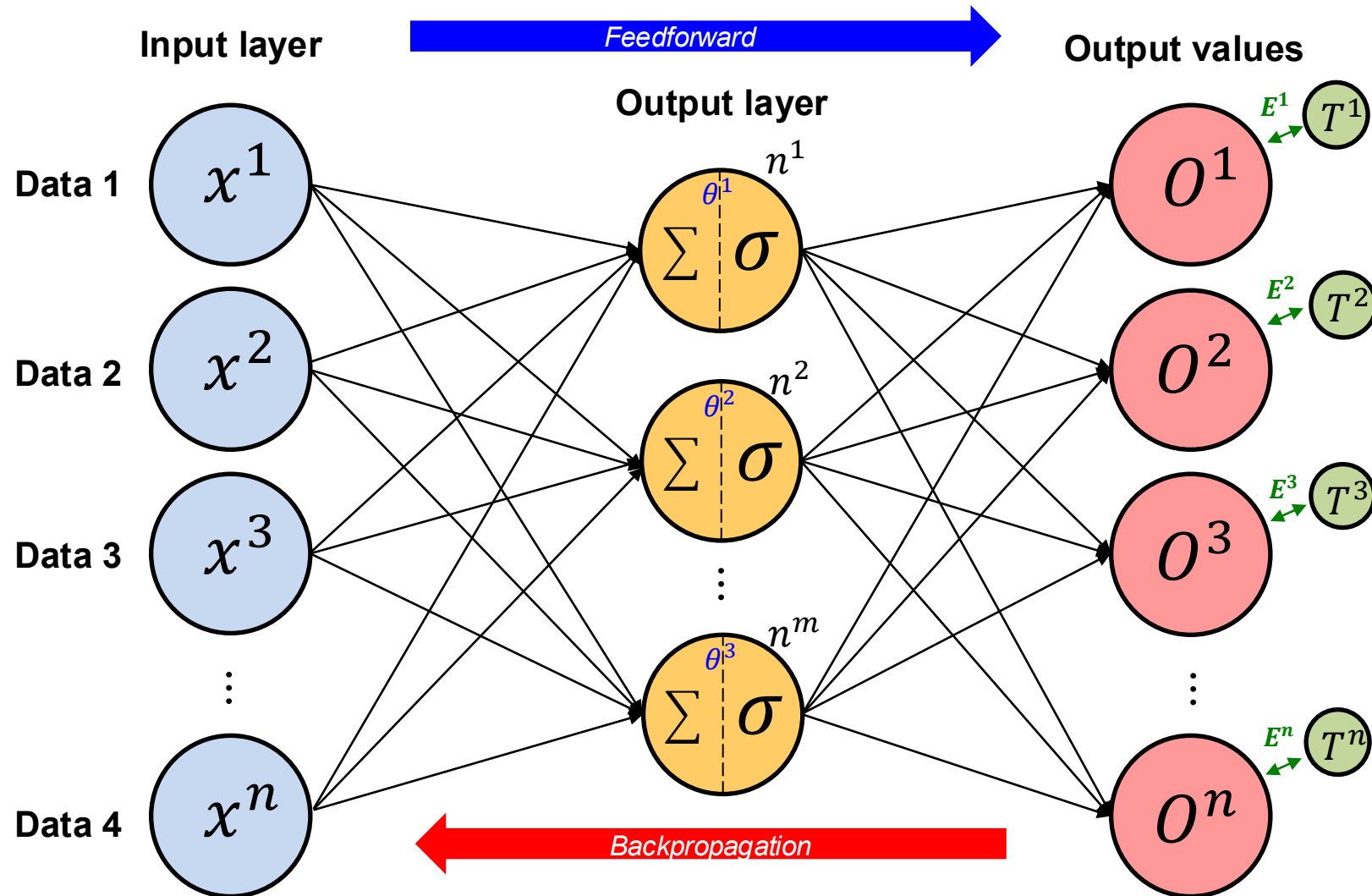
# Artificial Neural Network (ANN)



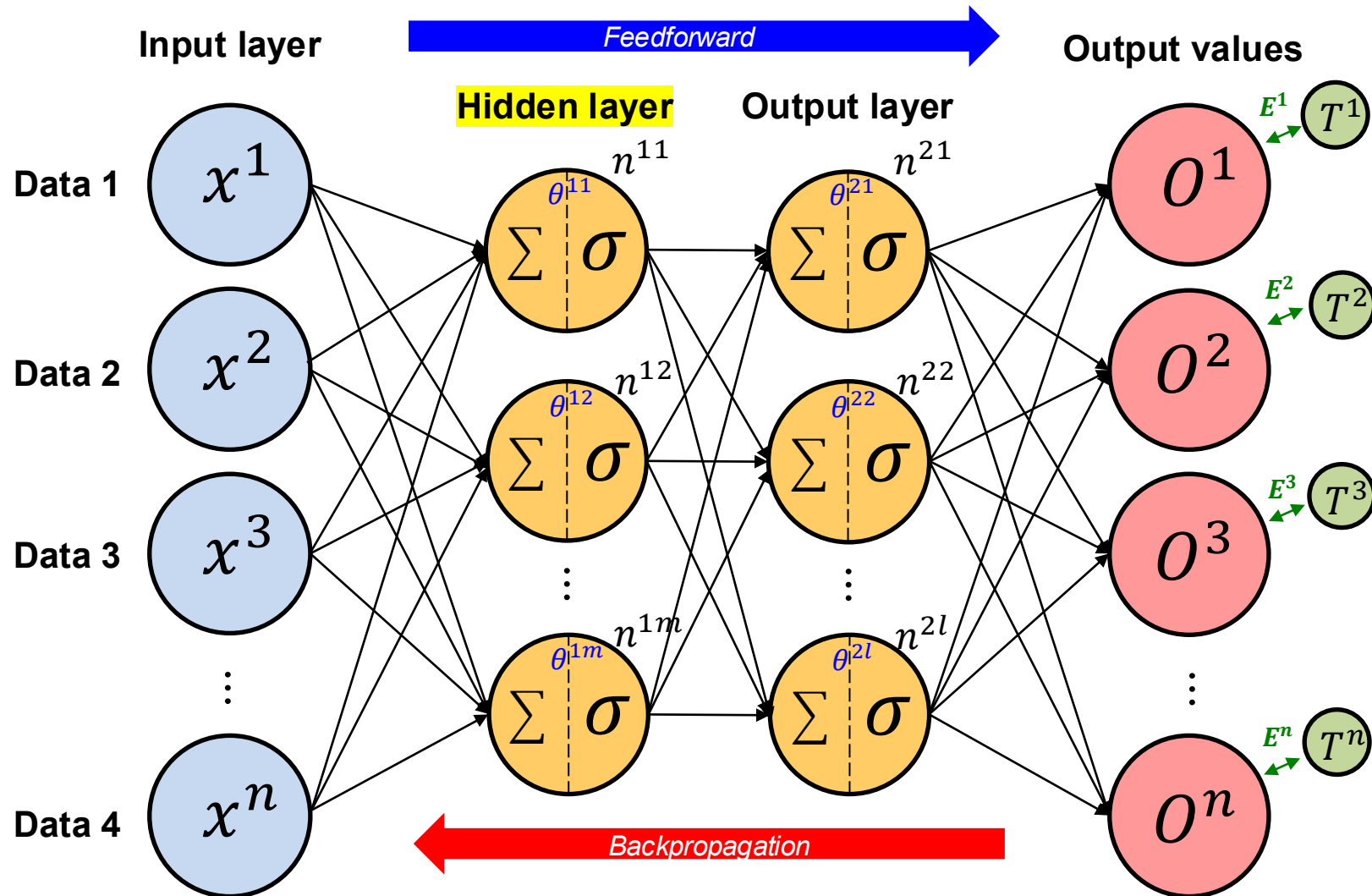
# Artificial Neural Network (ANN)



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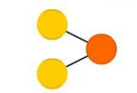
# Artificial Neural Network (ANN)

## A mostly complete chart of Neural Networks

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-  Backfed Input Cell
-  Input Cell
-  Noisy Input Cell
-  Hidden Cell
-  Probabilistic Hidden Cell
-  Spiking Hidden Cell
-  Output Cell
-  Match Input Output Cell
-  Recurrent Cell
-  Memory Cell
-  Different Memory Cell
-  Kernel
-  Convolution or Pool

Perceptron (P)



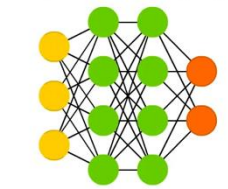
Feed Forward (FF)



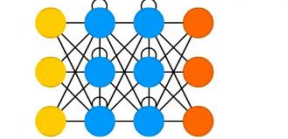
Radial Basis Network (RBF)



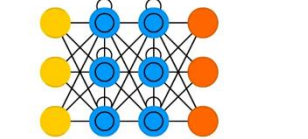
Deep Feed Forward (DFF)



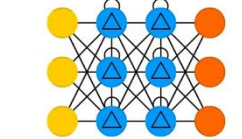
Recurrent Neural Network (RNN)



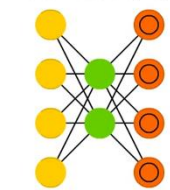
Long / Short Term Memory (LSTM)



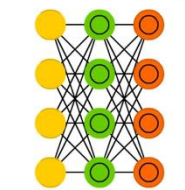
Gated Recurrent Unit (GRU)



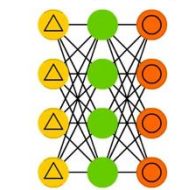
Auto Encoder (AE)



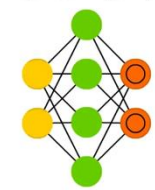
Variational AE (VAE)



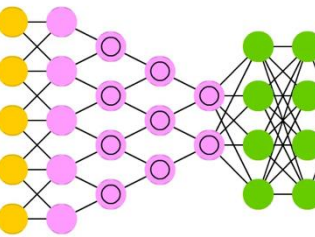
Denoising AE (DAE)



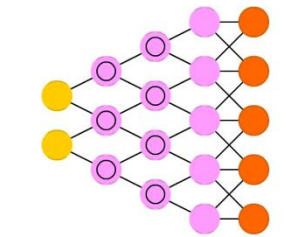
Sparse AE (SAE)



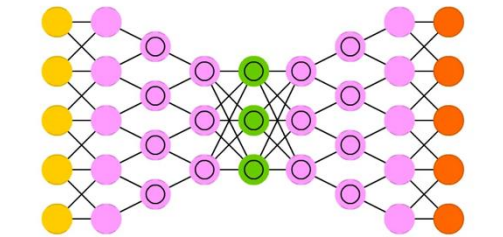
Deep Convolutional Network (DCN)



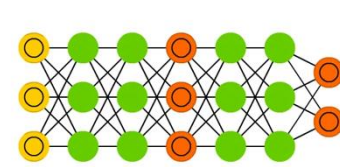
Deconvolutional Network (DN)



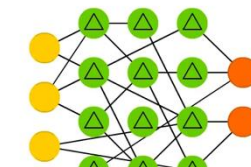
Deep Convolutional Inverse Graphics Network (DCIGN)



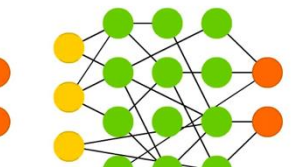
Generative Adversarial Network (GAN)



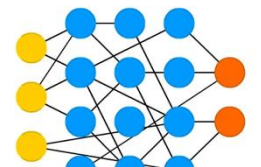
Liquid State Machine (LSM)



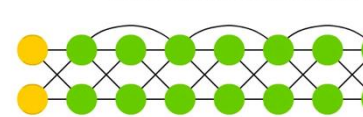
Extreme Learning Machine (ELM)



Echo State Network (ESN)



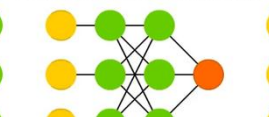
Deep Residual Network (DRN)



Kohonen Network (KN)



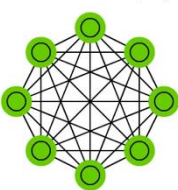
Support Vector Machine (SVM)



Neural Turing Machine (NTM)



Markov Chain (MC)



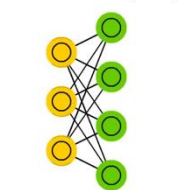
Hopfield Network (HN)



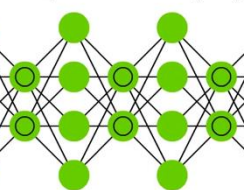
Boltzmann Machine (BM)



Restricted BM (RBM)



Deep Belief Network (DBN)

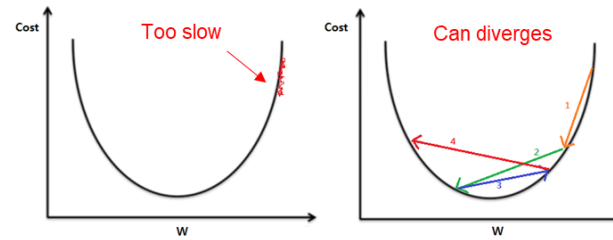


- Key hyperparameters of ANN

1) Number of neurons in each hidden layer (+ number of layers) and number of epochs for training

→ Generally, the number of neurons is increased as the dimensionality of the data increases, but it is not always the best solution.

2) Learning rate  $\alpha$  ( $\omega = \omega + \alpha \Delta \omega$ ,  $\theta = \theta + \alpha \Delta \theta$ )



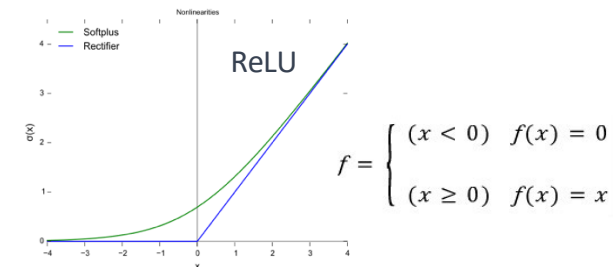
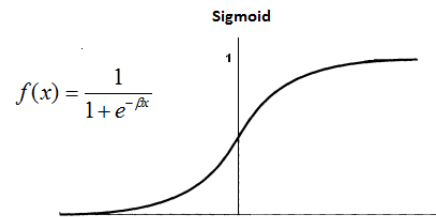
3) Activation function

→ Limitations of classic activation function 'sigmoid':

- The value is determined between 0 and 1 and changes abruptly in a specific range.
- As the operation becomes more complex (deep learning), the change in the gradient becomes negligible (Vanishing).

→ The most used activation function recently: ReLU (Rectified Linear Unit)

- Outputs 0 for inputs below a threshold, allowing for partial activation (Solves the problems of Gradient Vanishing and Exploding)
- Very simple to calculate differentiation as it is a linear function



# Data Labeling for ANN (One-Hot Encoding)

Training Dataset (80%)

**288** rows X **30** columns

Training Label

**288** rows X **2** columns

Test Dataset (20%)

**72** rows X **30** columns

Test Label

**72** rows X **2** columns

